

MPLADS IntelliTrack

Complete Solution Blueprint & Architecture Specification | SIH Ready

1. Core Solution & Fundamental Principle

MPLADS IntelliTrack is an AI-powered monitoring and decision-support platform that creates a digital evidence trail for every MPLADS work, continuously analyzes project, financial, execution and compliance data, detects anomalies and emerging risks, predicts delays and cost overruns, and provides explainable alerts and recommendations to the appropriate authority for human verification and corrective action.

Fundamental Principle:
AI detects → Evidence supports → Authority verifies → Governance acts
(The system does not replace the existing MPLADS administrative process, sanction funds, or automatically declare fraud.)

2. Complete End-to-End Workflow

MP recommends work ↓ District Authority scrutinizes ↓ District Authority sanctions/ approves ↓ Implementing Agency assigned ↓ Tender / procurement ↓ Contractor selected ↓ Project execution ↓ Progress + expenditure + bills + evidence submitted ↓ Data validation ↓ AI / Analytics Engine ↓ Anomaly + Prediction + Compliance Analysis ↓ Risk Score ↓ Explainable Alert ↓ Responsible Authority ↓ Human / Field Verification ↓ Corrective Action / Closure ↓ Audit Trail

3. Core System Modules (10 Pillars)

Module 1 — Project Management

Project registration, MP recommendation, sanction info, unique project ID, location, category, cost, timeline, milestones, implementing agency, and contractor.

Module 2 — Digital Evidence

In-app camera capturing GPS, timestamp, project geofence, progress photographs, documents, bills, and inspection evidence.

Module 6 — Predictive Intelligence

Delay prediction, cost-overrun prediction, project risk trajectories, and contractor pattern analysis.

Module 7 — Risk Engine

Multi-factor risk score calculation, risk categories, severity classification, explainable AI, and historical tracking.

Module 3 — Financial Intelligence

Expenditure monitoring, cost benchmarking, bill analysis, invoice verification, material-rate comparison, duplicate invoice detection, and payment pattern analysis.

Module 4 — Execution Intelligence

Milestone tracking, progress monitoring, physical vs. financial progress evaluation, delay detection/prediction, and project health.

Module 5 — AI Anomaly Detection

Cost, financial, duplicate work, invoice, progress, evidence, and compliance anomaly identification engines.

Module 8 — Decision Support

Smart alerts, intelligent recommendations, structured verification workflow, and corrective-action tracking.

Module 9 — Governance

RBAC, jurisdiction control, immutable audit trails, data integrity mechanisms, and administrative monitoring.

Module 10 — Analytics & Dashboards

MoSPI, State, District, MP, and Agency dashboards, contractor workspace, GIS map, and national trends.

4. Data Architecture & 4-Layer Separation

To ensure government-grade auditability, information is strictly separated into four discrete layers:

Layer	Description	Examples
A. Source Data	Actual submitted/recorded information	Sanction amount, Invoice, Payment, Progress %, Location, Photograph
B. Derived Data	System calculations	Cost deviation %, Expenditure %, Delay days, Distance from location
C. AI Inference	Model-generated conclusions	Anomaly probability, Delay probability, Duplicate similarity, Risk score
D. Human Decision	Official administrative outcome	Verified, Not verified, Corrective action, Escalated, Closed, Under investigation

5. Database Design & Project ID Schema

Example Project ID: MPLADS-MH-AMR-2026-00452

```

USER |— Role |— State |— District |— Jurisdiction
MP |— Constituency
PROJECT |— Recommendation |— Sanction |— Location |— Milestones |— Implementing
Agency |— Contractor
PROJECT |— EXPENDITURE |— INVOICE |— EVIDENCE |— PROGRESS |— INSPECTION |—
ANOMALY |— RISK_SCORE |— ALERT |— AUDIT_LOG

```

Every related record links back to the Project ID, establishing a seamless traceable chain from Recommendation to Audit Log.

6. AI / ML Methodology & Deep Dives

7.1 Cost Anomaly Detection

Inputs: Project type, cost, capacity, location, material, labor, transportation, specifications, historical comparables.

Technique: Statistical benchmarking + regression/outlier detection (Isolation Forest / XGBoost).

Output Example: Expected Cost: ₹43.2L | Actual Estimate: ₹50L | Deviation: +15.7% | **Cost Risk: HIGH**

8. Progress – Expenditure Mismatch

Formula: Financial Progress = Expenditure / Sanctioned Amount. Compared dynamically against physical progress milestones. *Example: Physical = 42%, Financial = 83% → Mismatch = 41% → High-risk alert.*

9. Delay Prediction & 10. Duplicate Project Detection

Delay Prediction (XGBoost/Random Forest): Predicts probability of delay based on velocity, milestone adherence, and expenditure rate.

Duplicate Detection: Uses NLP (Sentence Transformers) for description text similarity + Geospatial coordinate matching + Structured attributes (cost, type, time overlap).

11. Invoice Intelligence Pipeline

Invoice Image/PDF → OCR → Document Parsing → Structured Data → BOQ/Estimate Comparison → Historical Benchmark → Anomaly Flag.

7. Risk Engine, Explainability & Smart Alerts

Risk Levels

- 0–30 Normal** No significant anomaly.
- 31–50 Watch** Minor deviation.
- 51–75 Review** Requires closer examination.
- 76–100 High Risk** Prioritized human verification.

Explainable AI (XAI) Example

Risk Score: 76 (High)

- Cost is 18% above comparable projects.
- Financial progress (78%) exceeds physical progress (46%).
- Execution velocity indicates probable delay.

Recommendation: Verify BOQ, expenditure docs, and physical progress.

8. Project Health Card & GIS Intelligence

PROJECT HEALTH CARD (Sample)

Project: Community Water Tank | **ID:** MPLADS-MH-AMR-2026-00452

Sanctioned: ₹50L | **Expenditure:** ₹37L | **Physical:** 48% | **Financial:** 74%

Timeline: Month 8 / 10 expected | **Risk Score:** **78/100 HIGH RISK**

Flags: ⚠ Cost deviation | ⚠ Progress-expenditure mismatch | ⚠ Delay probability: 72%

Evidence: ✓ 14 photos (GPS verified) | ✓ 5 invoices | ⚠ 1 invoice anomaly

Recommended Action: → Physical field verification & expenditure review.

9. Prototype Scope & SIH Execution Strategy

For the SIH prototype, we execute a complete vertical slice rather than sprawling features:

1. Create MPLADS project & assign agency/contractor.
2. Contractor submits progress, photo, GPS, and invoice.
3. System validates evidence and runs multi-model AI analysis.
4. Demonstrate 5 anomalies (Cost, Progress/Expenditure, Delay, Duplicate, Invoice).
5. Generate Risk Score (84 - High Risk), explanation, and route smart alert to District Authority.
6. Officer records verification and audit trail records all steps.

10. Technology Stack & Architecture

Tier	Technology
Frontend	React.js / Next.js (Dashboards & role-specific interfaces)
Backend	Java Spring Boot (Enterprise architecture & core APIs)
AI / ML Services	Python FastAPI + Scikit-learn, XGBoost, OpenCV, Sentence Transformers, SHAP
Database & Storage	PostgreSQL with PostGIS extension + Secure Object Storage

11. Final USP & Expected Impact

Final SIH Presentation USP

“MPLADS IntelliTrack transforms MPLADS monitoring from periodic manual review into continuous, evidence-driven risk intelligence.”

It connects every project with its financial records, execution evidence, and administrative history, uses AI to detect anomalies and predict risks, explains why a project is flagged, and routes the case to the responsible authority for verification and action.